

Tutorial III – Stability

Longitudinal stability describes the aircraft's response to a perturbation from its trimmed state, e.g. a change in angle of attack. Steady horizontal flight requires the aircraft to be trimmed, in order to eliminate pitching moments. This condition is achieved by deflection of control surfaces and adjustment of thrust. These characteristics are affected by the aircraft configuration, center of gravity location and compressibility effects.



Basics


 Derive the equation for the neutral point. Start with the moment equilibrium of the aircraft neglecting fuselage lift, drag and propulsion.

Calculation tasks

- 1) Determine the mean aerodynamic chord c_{MAC} of the Airbus A320. Apply the algebraic method and the geometric method (approximate the wing with a trapezoidal shape to be able to use these methods). Compare the results with the manufacturer's value.

The following tasks (2-5) are carried out with the goal to calculate the aircraft's neutral point.

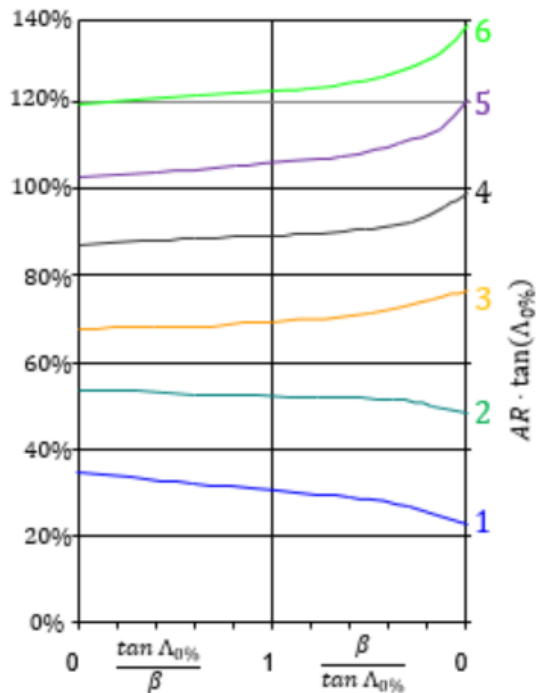
- 2) Calculate the lift coefficient slope $C_{L\alpha,j}$ of each aircraft assembly (wing, horizontal tail and fuselage) at $M = 0.78$. Scale and correct $C_{L\alpha,HT}$ to aircraft level.
- 3) Determine the normalized aerodynamic center of the main wing $\bar{x}_{AC,W}$ and the horizontal tail $\bar{x}_{AC,HT}$. The fixed distance from the nose of the aircraft to each apex can be found in the aircraft dataset.
- 4) Determine the moment coefficient slope $C_{M\alpha,fus}$ of the fuselage.
- 5) Finally, locate the aircraft's aerodynamic center $\bar{x}_{AC}$ and compute the stability measure, assuming that the center of gravity location x_{CG} in cruise is at 90% of root chord.
- 6) Determine the horizontal tail volume V_{HT} of the aircraft. Compare your result with the horizontal tail volume of similar aircraft.

Assumptions and Additional Data

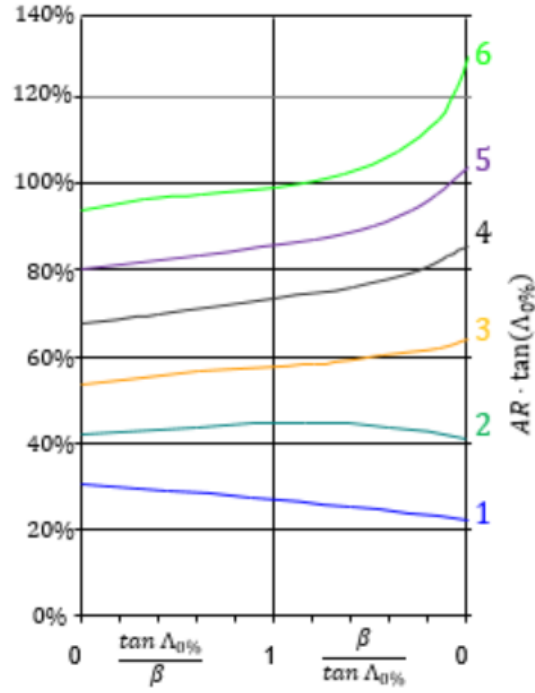
Flow conditions at horizontal tail:

- Ratio of dynamic pressure at horizontal tail to aircraft: $q_{HT}/q = 0.85$
- Ratio of effective angle of attack at horizontal tail to aircraft: $\partial\alpha_{HT}/\partial\alpha = 0.65$

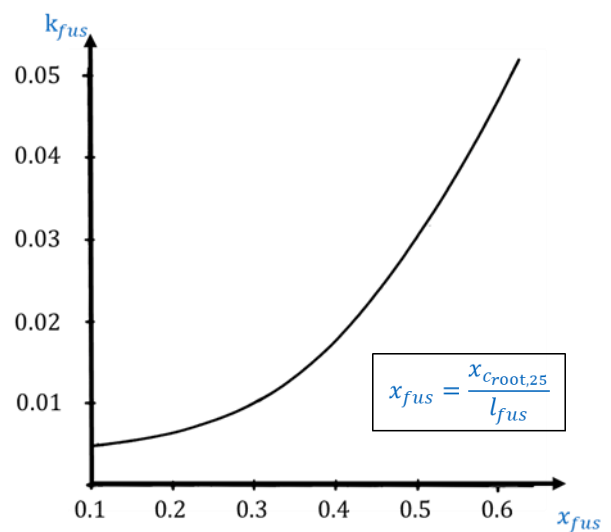
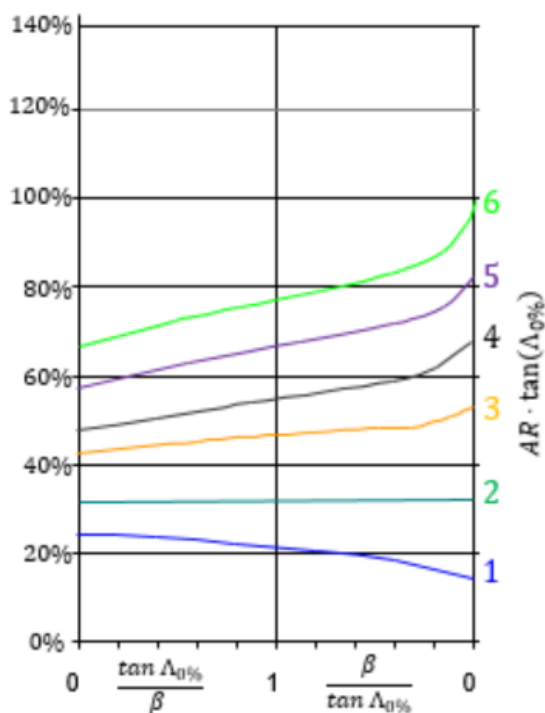
$\lambda = 0.5$



$\lambda = 0.25$



$\lambda = 0$



Equations

$$K = 1 + 0.75 \cdot \left(\frac{t}{c}\right)$$

$$\tan(\Lambda_{m\%}) = \tan(\Lambda_{n\%}) - \frac{4}{AR} \cdot \left[\frac{m-n}{100} \cdot \frac{1-\lambda}{1+\lambda} \right]$$

$$C_{M\alpha, fus} = 45 \cdot \frac{k_{fus} \cdot w_{fus}^2 \cdot l_{fus}}{c_{MAC} \cdot S_{ref}}$$

$$V_{HT} = \frac{S_{HT}}{S_{ref}} \cdot \frac{x_{AC, HT} - x_{AC, W}}{c_{MAC}}$$